SOC-5811 Week 7: Regression with multiple variables

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Review categorical predictor variables

```r
penguin_df <-
  palmerpenguins::penguins %>%
  na.omit()
head(penguin_df)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           36.7          19.3               193        3450
## 5 Adelie  Torgersen           39.3          20.6               190        3650
## 6 Adelie  Torgersen           38.9          17.8               181        3625
## # i 2 more variables: sex <fct>, year <int>
```

Review categorical predictor variables

What happens if we use a categorical predictor variable with **two** levels?

$$y_i = \beta_0 + \beta_1(\text{Sex})$$

Review categorical predictor variables

```r
summary(lm(bill_depth_mm ~ sex, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.4255     0.1425 115.286  < 2e-16 ***
## sexmale       1.4656     0.2006   7.307 2.07e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
## Multiple R-squared:  0.1389, Adjusted R-squared:  0.1363
## F-statistic: 53.39 on 1 and 331 DF,  p-value: 2.066e-12
```

Review categorical predictor variables

```
penguin_df <- penguin_df %>%
 dummy_cols('sex')
penguin_df %>%
  select(species,bill_length_mm,bill_depth_mm,sex,sex_female,sex_male)
```

```
## # A tibble: 333 x 6
##    species bill_length_mm bill_depth_mm sex    sex_female sex_male
##    <fct>            <dbl>         <dbl> <fct>       <int>    <int>
##  1 Adelie            39.1          18.7 male            0        1
##  2 Adelie            39.5          17.4 female          1        0
##  3 Adelie            40.3          18   female          1        0
##  4 Adelie            36.7          19.3 female          1        0
##  5 Adelie            39.3          20.6 male            0        1
##  6 Adelie            38.9          17.8 female          1        0
##  7 Adelie            39.2          19.6 male            0        1
##  8 Adelie            41.1          17.6 female          1        0
##  9 Adelie            38.6          21.2 male            0        1
## 10 Adelie            34.6          21.1 male            0        1
## # i 323 more rows
```

Review categorical predictor variables

```r
summary(lm(bill_depth_mm ~ sex_male, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex_male, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.4255     0.1425 115.286  < 2e-16 ***
## sex_male      1.4656     0.2006   7.307 2.07e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
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```

Review categorical predictor variables

- Do I need to add both dummy variables to my regression model? Why or why not?

```
penguin_df %>%
  select(species,bill_length_mm,bill_depth_mm,sex,sex_female,sex_male)
```

```
## # A tibble: 333 x 6
##    species bill_length_mm bill_depth_mm sex    sex_female sex_male
##    <fct>            <dbl>         <dbl> <fct>       <int>    <int>
##  1 Adelie            39.1          18.7 male            0        1
##  2 Adelie            39.5          17.4 female          1        0
##  3 Adelie            40.3          18   female          1        0
##  4 Adelie            36.7          19.3 female          1        0
##  5 Adelie            39.3          20.6 male            0        1
##  6 Adelie            38.9          17.8 female          1        0
##  7 Adelie            39.2          19.6 male            0        1
##  8 Adelie            41.1          17.6 female          1        0
##  9 Adelie            38.6          21.2 male            0        1
## 10 Adelie            34.6          21.1 male            0        1
## # i 323 more rows
```

Review categorical predictor variables

```r
summary(lm(bill_depth_mm ~ sex_male + sex_female, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex_male + sex_female, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients: (1 not defined because of singularities)
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.4255     0.1425 115.286  < 2e-16 ***
## sex_male      1.4656     0.2006   7.307 2.07e-12 ***
## sex_female        NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
## Multiple R-squared:  0.1389, Adjusted R-squared:  0.1363
## F-statistic: 53.39 on 1 and 331 DF,  p-value: 2.066e-12
```

Review multicollinearity

- The problem of **multicollinearity**.

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- The problem of **multicollinearity**.
- In a regression model, you can't have predictors that are a perfect function of one another. In other words, you can't have perfect **multicollinearity** where predictors are perfectly correlated with one another.

- The problem of **multicollinearity**.
- In a regression model, you can't have predictors that are a perfect function of one another. In other words, you can't have perfect **multicollinearity** where predictors are perfectly correlated with one another.
- Why?

Review

1. Choose a functional form for the model.

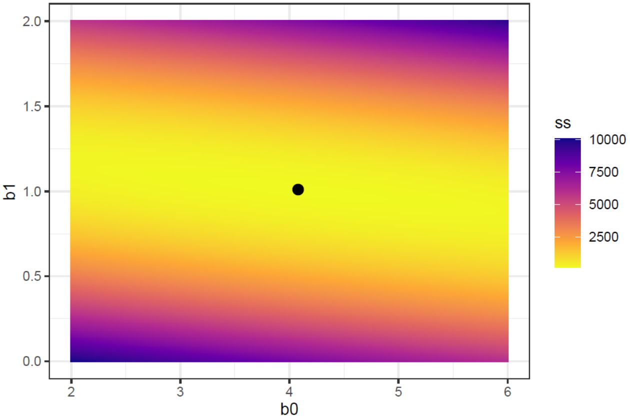
$$\text{depth}_i = \beta_0 + \beta_1 \text{sex\_male} + \beta_2 \text{sex\_female} + \epsilon_i$$

2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).

$$\text{Sum of squared residuals} = \sum_{i=1}^{n} (y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i)^2$$

3. Fit the model (e.g., estimate parameters) by optimizing the objective function.

Review multicollinearity



- If you provide a categorical predictor variable to lm(), the default behavior is to transform it into N-1 dummy variables where N is the number of unique levels.

Review categorical predictor variables

```r
penguin_df <- penguin_df %>%
 dummy_cols('species')
head(penguin_df)
```

```
## # A tibble: 6 x 13
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           36.7          19.3               193        3450
## 5 Adelie  Torgersen           39.3          20.6               190        3650
## 6 Adelie  Torgersen           38.9          17.8               181        3625
## # i 7 more variables: sex <fct>, year <int>, sex_female <int>, sex_male <int>,
## #   species_Adelie <int>, species_Chinstrap <int>, species_Gentoo <int>
```

Review categorical predictor variables

```r
summary(lm(bill_depth_mm ~ species_Adelie + species_Gentoo + species_Chinstrap,
           data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ species_Adelie + species_Gentoo +
##     species_Chinstrap, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.84726 -0.79664  0.00336  0.75274  3.15274
##
## Coefficients: (1 not defined because of singularities)
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       18.42059    0.13627 135.181   <2e-16 ***
## species_Adelie    -0.07333    0.16497  -0.444    0.657
## species_Gentoo    -3.42395    0.17082 -20.044   <2e-16 ***
## species_Chinstrap       NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.124 on 330 degrees of freedom
## Multiple R-squared:  0.6764, Adjusted R-squared:  0.6744
## F-statistic: 344.8 on 2 and 330 DF,  p-value: < 2.2e-16
```

Review categorical predictor variables

```r
penguin_df <- penguin_df %>%
  mutate(species = factor(species, levels=c('Chinstrap','Adelie','Gentoo')))
summary(lm(bill_depth_mm ~ species, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ species, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.84726 -0.79664  0.00336  0.75274  3.15274
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   18.42059    0.13627 135.181   <2e-16 ***
## speciesAdelie -0.07333    0.16497  -0.444    0.657
## speciesGentoo -3.42395    0.17082 -20.044   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
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Review categorical predictor variables

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

Review categorical predictor variables

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$

Review categorical predictor variables

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$

$$E[Y|\text{species} = \text{Adelie}] = (18.42)+(-0.07)(1)+(-3.42)(0) = \mathbf{18.35}$$

Review categorical predictor variables

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$

$$E[Y|\text{species} = \text{Adelie}] = (18.42)+(-0.07)(1)+(-3.42)(0) = \mathbf{18.35}$$

$$E[Y|\text{species} = \text{Chinstrap}] = (18.42)+(-0.07)(0)+(-3.42)(0) = \mathbf{18.42}$$

Multiple predictors

- The coefficient β_k is the average or expected difference in outcome y, comparing two people who differ by one unit in the predictor x_k while being equal in all the other predictors. This is sometimes stated in shorthand as comparing two people (or, more generally, two observational units) that differ in x_k with all the other predictors held constant.

Multiple predictors

```r
mod <- lm(bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
summary(mod)
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.63218    0.87109   9.910  < 2e-16 ***
## bill_length_mm   0.20044    0.01768  11.337  < 2e-16 ***
## speciesAdelie    1.93308    0.22572   8.564 4.26e-16 ***
## speciesGentoo   -3.17024    0.14679 -21.598  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9543 on 329 degrees of freedom
## Multiple R-squared:  0.7673, Adjusted R-squared:  0.7652
## F-statistic: 361.6 on 3 and 329 DF,  p-value: < 2.2e-16
```

Multiple predictors

- $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63.**

Multiple predictors

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Multiple predictors

- $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63.**
- $\beta_1 = 0.20$: Comparing penguins within the same species but whose bill lengths differ by 1, we would predict their difference in bill depths to be **0.20.**
- $\beta_2 = 1.93$: Comparing penguins who have the same bill length but one is Chinstrap and one is Adelie, we would predict their difference in bill depths to be **1.93.**

Multiple predictors

- $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63.**
- $\beta_1 = 0.20$: Comparing penguins within the same species but whose bill lengths differ by 1, we would predict their difference in bill depths to be **0.20.**
- $\beta_2 = 1.93$: Comparing penguins who have the same bill length but one is Chinstrap and one is Adelie, we would predict their difference in bill depths to be **1.93.**
- $\beta_3 = -3.17$: Comparing penguins who have the same bill length but one is Chinstrap and one is Gentoo, we would predict their difference in bill depths to be **3.17.**

Multiple predictors

What are we interpreting?

- Regression is a way of making comparisons; it is very important to understand and be able to articulate the comparison you are making.

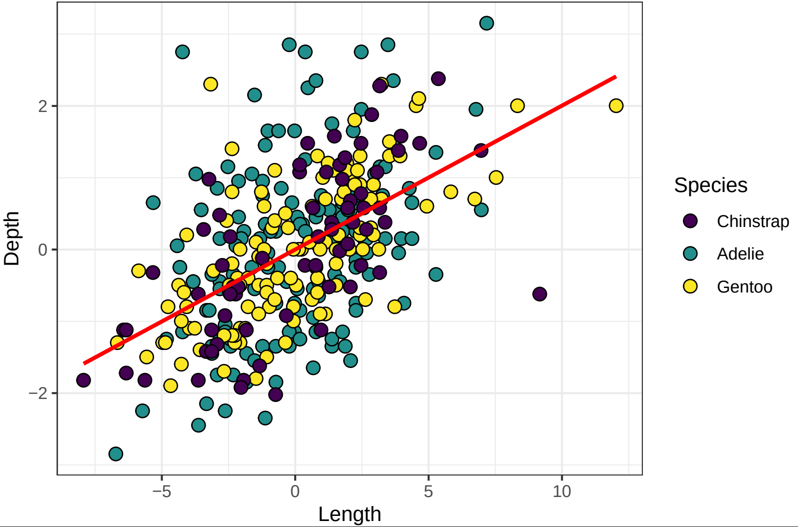
$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

$$\beta_1 = E[d|l = x + 1, s] - E[d|l = x, s] = \mathbf{0.20}$$

$$\beta_2 = E[d|s = \text{Adelie}, l] - E[d|s = \text{Chinstrap}, l] = \mathbf{1.93}$$

$$\beta_3 = E[d|s = \text{Gentoo}, l] - E[d|s = \text{Chinstrap}, l] = \mathbf{-3.17}$$

Holding something constant

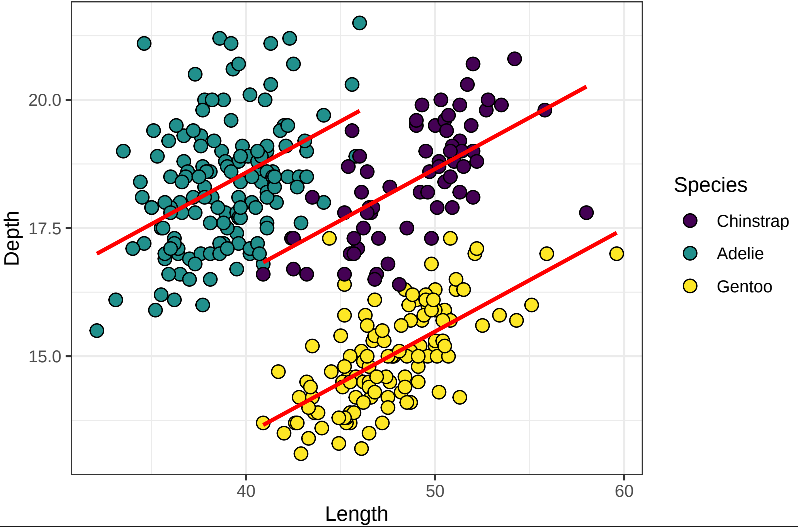


Multiple predictors

```r
mod <- lm(bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
predictions <- tibble(depth_preds = predictions(mod)$estimate)
penguin_df <- penguin_df %>%
  bind_cols(., predictions)
penguin_df %>% select(species,bill_length_mm,bill_depth_mm,depth_preds) %>% head()
```

```
## # A tibble: 6 x 4
## # Groups:   species [1]
##   species bill_length_mm bill_depth_mm depth_preds
##   <fct>            <dbl>         <dbl>       <dbl>
## 1 Adelie            39.1          18.7        18.4
## 2 Adelie            39.5          17.4        18.5
## 3 Adelie            40.3          18          18.6
## 4 Adelie            36.7          19.3        17.9
## 5 Adelie            39.3          20.6        18.4
## 6 Adelie            38.9          17.8        18.4
```

Multiple predictors



Interactions

- In our model, the slope of the regression for bill length was forced to be equal across all subgroups defined by species. In other words, we just have one β_1.

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i)$$

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

$$d_i = (l_i) + (s_i) + (l_i \times s_i)$$

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}}) + \beta_4(s_i^{\text{Adelie}} \times l_i) + \beta_5(s_i^{\text{Gentoo}} \times l_i)$$

Interactions

- No interaction:

$$d_i = l_i + s_i$$

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

- With interaction:

$$d_i = l_i + s_i + (l_i \times s_i)$$

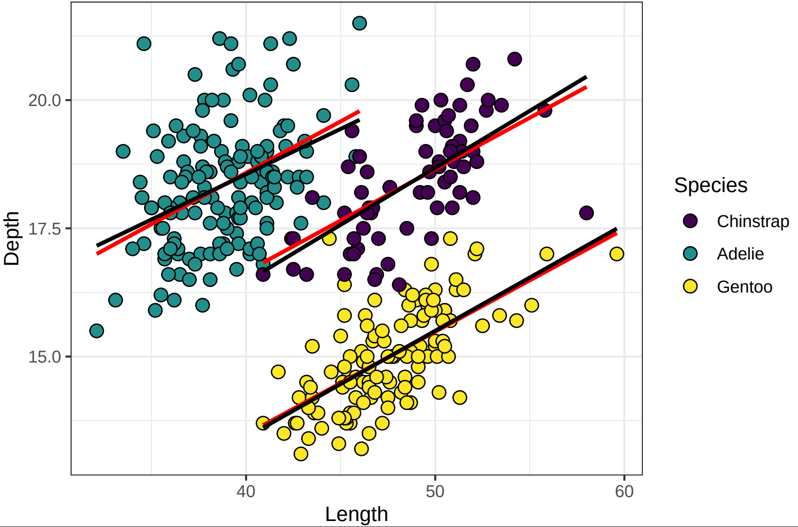
$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}}) + \beta_4(s_i^{\text{Adelie}} \times l_i) + \beta_5(s_i^{\text{Gentoo}} \times l_i)$$

Interactions

```r
mod2 <- lm(bill_depth_mm ~ bill_length_mm * species, data = penguin_df)
summary(mod2)
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm * species, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.6574 -0.6559 -0.0483  0.5203  3.4990
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    7.56914    1.71127   4.423 1.33e-05 ***
## bill_length_mm                 0.22221    0.03496   6.356 6.96e-10 ***
## speciesAdelie                  3.91857    2.06731   1.895   0.0589 .
## speciesGentoo                 -2.44818    2.17972  -1.123   0.2622
## bill_length_mm:speciesAdelie -0.04553    0.04594  -0.991   0.3224
## bill_length_mm:speciesGentoo -0.01460    0.04499  -0.324   0.7458
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9556 on 327 degrees of freedom
## Multiple R-squared:  0.7681, Adjusted R-squared:  0.7645
## F-statistic: 216.6 on 5 and 327 DF,  p-value: < 2.2e-16
```

Interactions



INTERACTIONS

```r
mod3 <- lm(bill_depth_mm ~ bill_length_mm * body_mass_g, data = penguin_df)
summary(mod3)
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm * body_mass_g, data = penguin_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.8374 -1.1902 -0.0181  1.2037  4.1935
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 1.367e+01  4.411e+00   3.099  0.00211 **
## bill_length_mm              1.920e-01  9.640e-02   1.992  0.04720 *
## body_mass_g                 7.061e-04  1.134e-03   0.623  0.53384
## bill_length_mm:body_mass_g -4.222e-05  2.409e-05  -1.752  0.08067 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.732 on 329 degrees of freedom
## Multiple R-squared:  0.2337, Adjusted R-squared:  0.2267
## F-statistic: 33.45 on 3 and 329 DF,  p-value: < 2.2e-16
```

Review: Why multiple variables?

- Social process are typically not the function of one independent variable

Review: Why multiple variables?

- Social process are typically not the function of one independent variable
- Simpsons paradox

Review: Why multiple variables?

- Social process are typically not the function of one independent variable
- Simpsons paradox
- "Holding constant" = "Comparing two items i and j that differ by an amount x on predictor k but are identical on all other predictors, the predicted difference $y_i - y_j$ is $B_k(x)$, on average."

Review: Why multiple variables?

Different research questions:

- What is the average bill depth of Chinstrap penguins?

Review: Why multiple variables?

Different research questions:

- What is the average bill depth of Chinstrap penguins?
- What is the difference in average bill depth between Chinstrap and Adelie penguins?

Review: Why multiple variables?

Different research questions:

- What is the average bill depth of Chinstrap penguins?
- What is the difference in average bill depth between Chinstrap and Adelie penguins?
- What is the difference in average bill depth between Chinstrap and Adelie penguins who have the same bill length?

Review: Why multiple variables?

Different research questions:

- What is the average bill depth of Chinstrap penguins?
- What is the difference in average bill depth between Chinstrap and Adelie penguins?
- What is the difference in average bill depth between Chinstrap and Adelie penguins who have the same bill length?
- What is the difference in average bill depth between penguins of the same species who differ in length by 1?

Confounding

- Review Simpson's Paradox

Department	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	**825**	62%	108	82%
B	585	63%	**560**	63%	25	68%
C	918	35%	325	37%	**593**	34%
D	792	34%	417	33%	375	35%
E	584	25%	191	28%	**393**	24%
F	714	6%	373	6%	341	7%
Total	**4526**	**39%**	**2691**	**45%**	**1835**	**30%**

Legend:

greater percentage of successful applicants than the other gender

greater number of applicants than the other gender

bold - the two 'most applied for' departments for each gender

Confounding

$$\text{Admission} = \beta_0 + \beta_1(\text{Gender})$$

$$\text{Admission} = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Department})$$

Confounding

- Do not listen to anyone who says "Well, you *have* to control for X."

Confounding

- Do not listen to anyone who says "Well, you *have* to control for X."
- What they are really saying: "The comparison in which you are not controlling for X is not useful; I'm assuming you want to make the comparison where you control for X." This can be an argument on theoretical grounds about the utility of various research questions.

Confounding

- Do not listen to anyone who says "Well, you *have* to control for X."
- What they are really saying: "The comparison in which you are not controlling for X is not useful; I'm assuming you want to make the comparison where you control for X." This can be an argument on theoretical grounds about the utility of various research questions.
- There may be very good reasons to control for X! But it *completely* depends on your research question; more specifically, exactly what **comparison** you say you're trying to make.

Confounding

- Do not listen to anyone who says "Well, you *have* to control for X."
- What they are really saying: "The comparison in which you are not controlling for X is not useful; I'm assuming you want to make the comparison where you control for X." This can be an argument on theoretical grounds about the utility of various research questions.
- There may be very good reasons to control for X! But it *completely* depends on your research question; more specifically, exactly what **comparison** you say you're trying to make.
- Transition to causal inference and the idea of *confounding*.